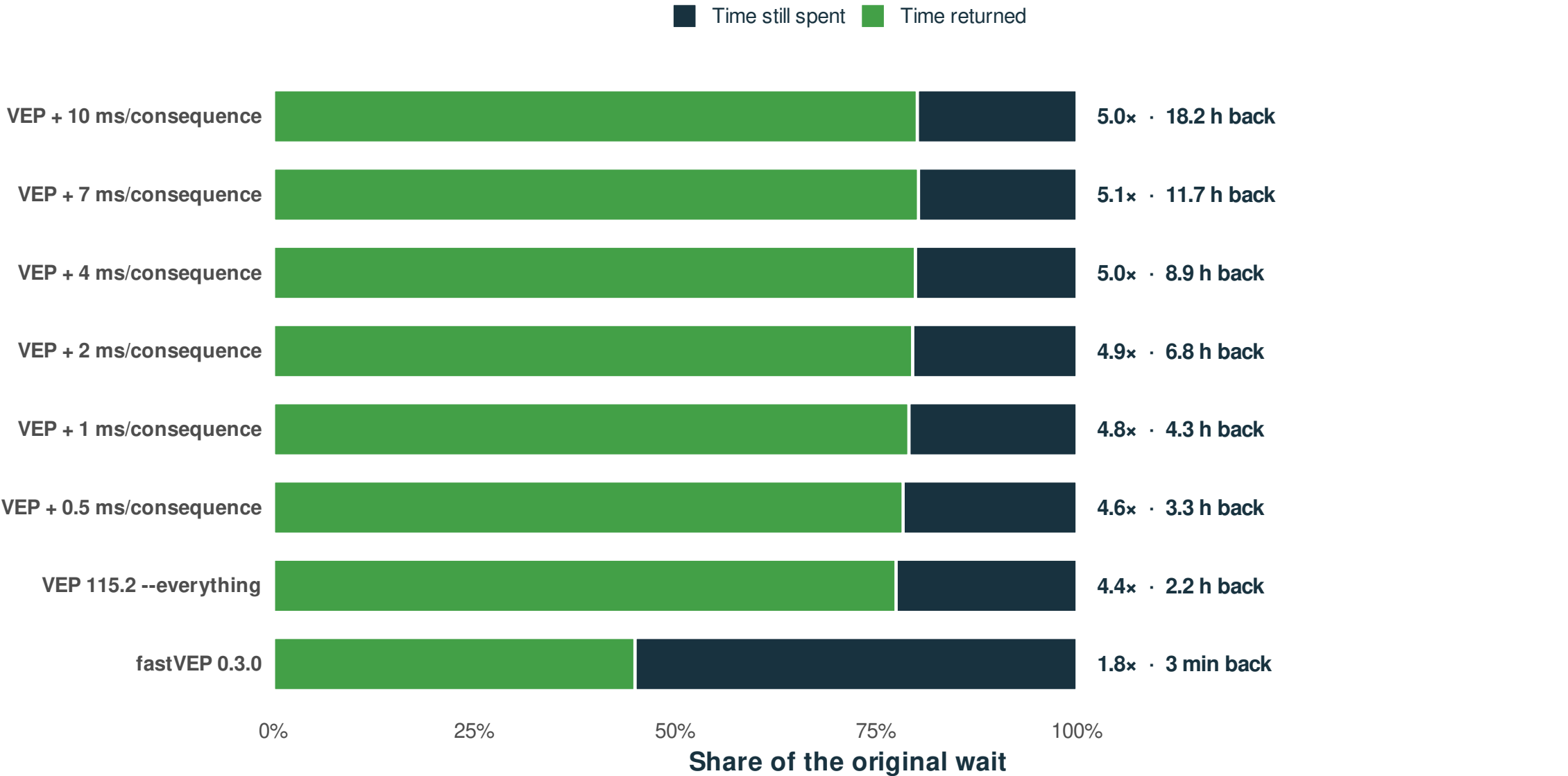


From fastVEP to an all-day pipeline: cache hits keep returning time

The same real WGS and approximately 80% reuse across the complete measured spectrum



Bars are normalized within each measured example; labels report measured speedup and absolute time returned. Every row uses KPGP-00319 (4.80M variants) and the AF $\geq 10\%$ cache. The synthetic plugin emits no fields; it changes pipeline cost without changing annotations.